

On Oversquashing in Graph Neural Networks Through the Lens of Dynamical Systems

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Generative and Deep Learning Course 2025/2026 - University of Pisa

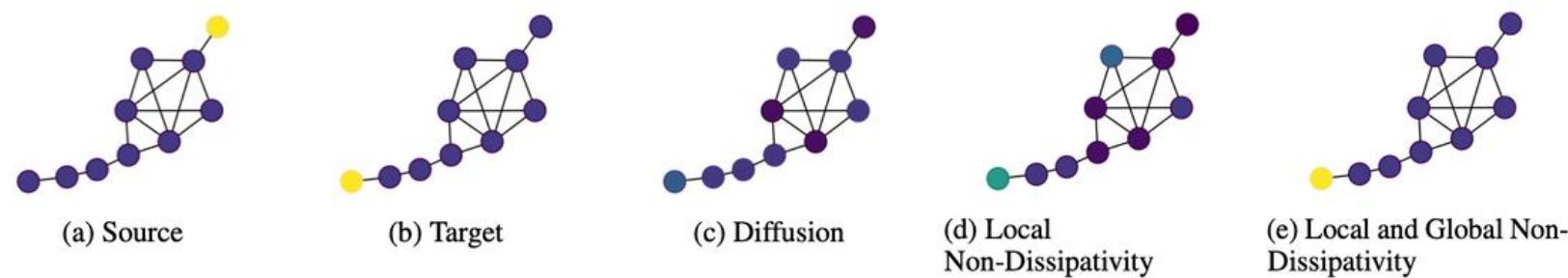


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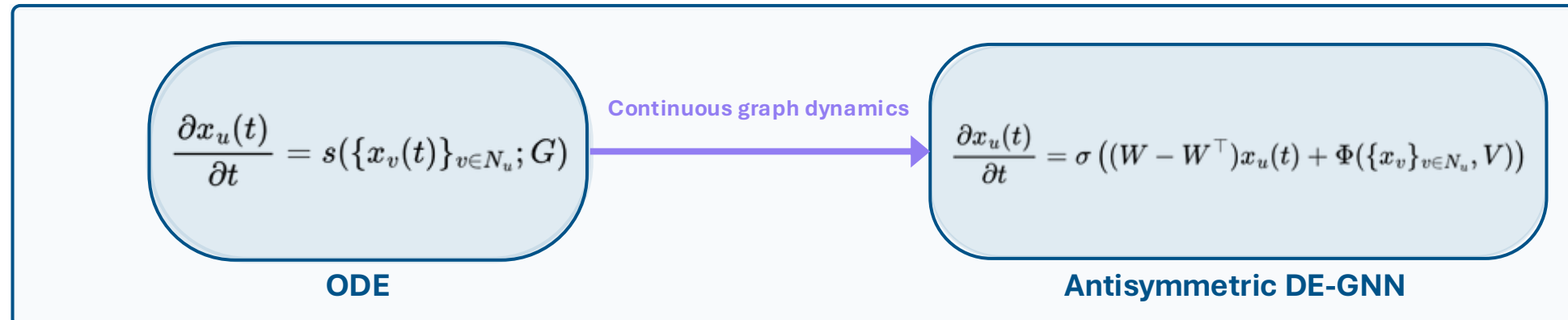
1. Introduction

Oversquashing in Graph Neural Networks

Graph Neural Networks propagate information through iterative message passing between neighboring nodes. However, distant-node information is compressed through graph bottlenecks, causing oversquashing and limiting long-range propagation.



ODE interpretation



Information Propagation Dynamics

The real part of the Jacobian eigenvalues determines the stability of information propagation.

Unstable $\max_i \operatorname{Re}(\lambda_i(J)) > 0$	Dissipative $\max_i \operatorname{Re}(\lambda_i(J)) < 0$	Non-Dissipative $\operatorname{Re}(\lambda_i(J)) = 0$
Sensitivity grows exponentially.	Sensitivity decays exponentially.	Sensitivity remains constant.
Effect: Exploding propagation. Small perturbations are amplified.	Effect: Information vanishes. Distant nodes lose influence (oversquashing).	Effect: Stable propagation. Information is preserved across long distances.

Key Insight: Dissipative dynamics lead to oversquashing, while non-dissipative dynamics (e.g., antisymmetric propagation in SWAN) preserve long-range information flow.

2. Approach

SWAN approach: antisymmetric graph dynamics

SWAN modifies message passing so information does not behave like pure diffusion, which usually smooths node features as depth increases.

Graph-wise SWAN used for global non-dissipativity

$$\partial X(t)/\partial t = \sigma(X(t)(W - W^T) + (\hat{A} + \hat{A}^T)X(t)(V - V^T) + \beta(\hat{A} - \hat{A}^T)X(t)(Z + Z^T))$$

1. Weight antisymmetry

$W - W^T$ stabilizes the node-wise transformation.

2. Spatial antisymmetry

$\hat{A} - \hat{A}^T$ makes the graph flow directional and non-diffusive.

3. Depth as time

t is the layer/depth variable:
 $X(0) \rightarrow X(t)$.

Global non-dissipative result

SWAN keeps the propagation rate constant:

$$\|\partial \operatorname{vec}(X(t)) / \partial \operatorname{vec}(X(0))\| = c$$

So increasing layers does not exponentially erase the input signal.

Why it matters

Diffusion GNNs instead decay with depth:

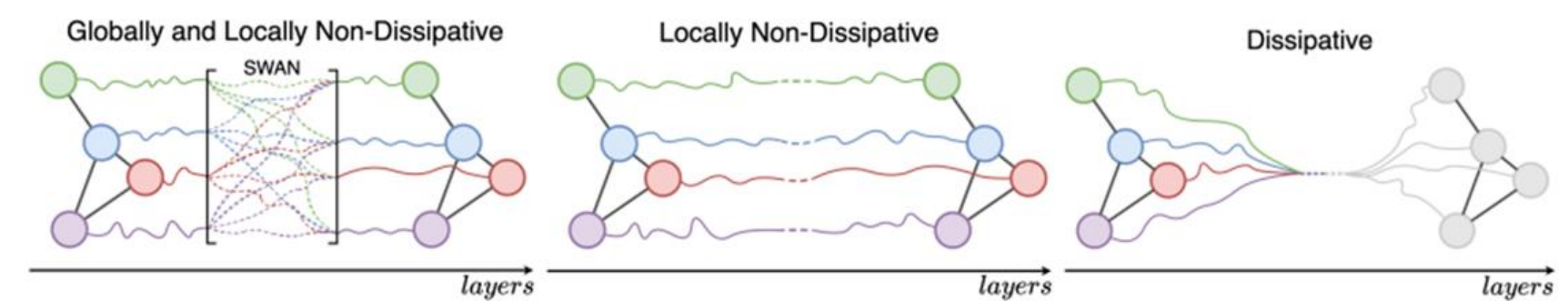
$$\|\partial \operatorname{vec}(X(t)) / \partial \operatorname{vec}(X(0))\| = |e^{(-tK)}|$$

This explains **oversquashing** / loss of long-range information.

OVERSQUASHING BOUND INTUITION

Bound goes from $\left\| \frac{\partial x_u(t)}{\partial x_u(0)} \right\| \leq \underbrace{(c_{\text{wpp}})^t}_{\text{model}} \underbrace{(\mathbf{O}^t)_{uu}}_{\text{topology}}$ to $\left\| \frac{\partial x_u(t)}{\partial x_u(0)} \right\| \leq \underbrace{(c_{\text{wpp}})^t}_{\text{model}} \underbrace{((c_{\text{w}} \mathbf{I} + c_{\text{a}} \mathbf{A} + \beta c_{\text{a}} \mathbf{S})^t)_{uu}}_{\text{topology}}$ mitigating the $\left\| \frac{\partial x_u(t)}{\partial x_u(0)} \right\| \rightarrow 0$ problem.

Take-home: SWAN adds antisymmetric graph dynamics, preserves global information flow, and strengthens long-range node sensitivity.



3. Experiments

Objectives:

- Can SWAN effectively propagate information to distant nodes?
- Can SWAN accurately predict graph properties related to long-range interactions?
- How does SWAN perform on real-world long-range benchmarks?

Models:

- Baselines:** MPNNs, DE-GNNs (such as ADGN) and oversquashing designated methods
- SWAN Variants:**
 - SWAN and SWAN-LEARN: pre-defined/learned aggregation terms $\hat{A}$ and $\tilde{A}$
 - SWAN-NE and SWAN-NE-LEARN: non-enforced variants of SWAN and SWAN-LEARN
 - SWAN $\beta=0$: a model with antisymmetry solely in the weight space

Datasets:

- Graph Transfer:** label transfer across k hops (3 topologies, 4 distances)
- Graph Property Prediction:** predicting Diameter, Single-Source Shortest Paths (SSSP) and Eccentricity on synthetic graphs
- Long-Range Graph Benchmark:** Peptides-func (AP), Peptides-struct (MAE) and PascalVOC-sp (F1) datasets

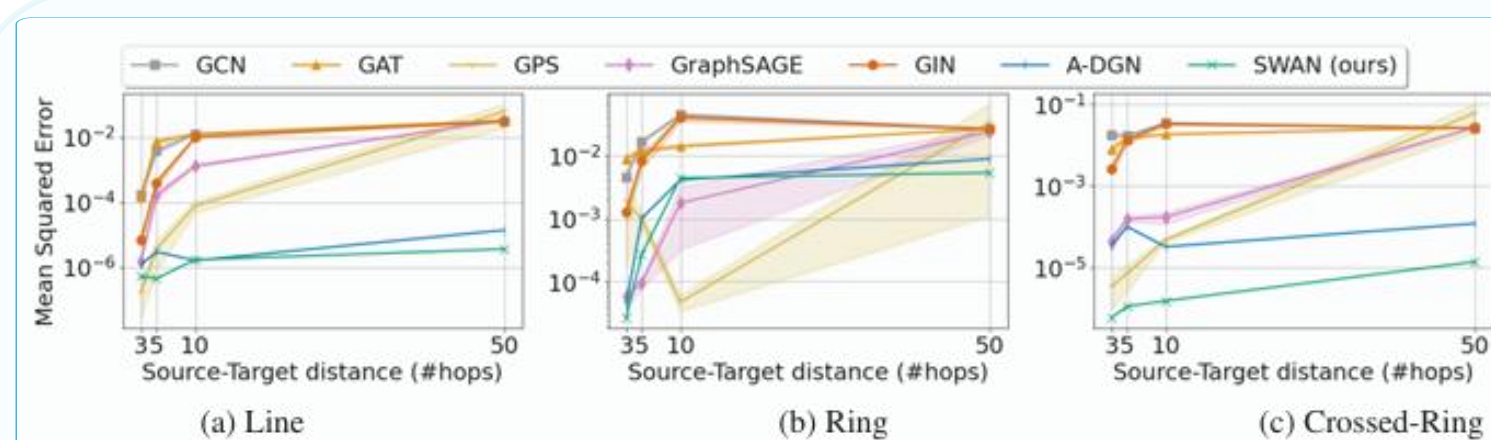


Figure 1: Information transfer performance on (a) Line, (b) Ring, and (c) Crossed-Ring graphs.

Table 2: Performance across three LRGB tasks. Results are averaged over 3 weight initializations. The first, second, and third best results for each task are color-coded.

	Model	Diameter	SSSP	Eccentricity
Table 1: Mean test set log10(MSE) and std averaged over 4 random weight initializations on the Graph Property Prediction tasks. The first, second, and third best results for each task are color-coded.	MPNNs			
	GCN	0.7424 $\pm$ 0.0466	0.9499 $\pm$ 0.18-10 ⁻⁵	0.8468 $\pm$ 0.0028
	GAT	0.8221 $\pm$ 0.0752	0.6951 $\pm$ 0.1499	0.7909 $\pm$ 0.0222
	GraphSAGE	0.8645 $\pm$ 0.0401	0.2863 $\pm$ 0.1843	0.7863 $\pm$ 0.0207
	GIN	0.6131 $\pm$ 0.0990	-0.5408 $\pm$ 0.4193	0.9504 $\pm$ 0.0007
	GCNII	0.5287 $\pm$ 0.0570	-1.1329 $\pm$ 0.0135	0.7640 $\pm$ 0.0355
Table 2: Performance across three LRGB tasks. Results are averaged over 3 weight initializations. The first, second, and third best results for each task are color-coded.	DE-GNNs			
	DGC	0.6028 $\pm$ 0.0050	-0.1483 $\pm$ 0.0231	0.8261 $\pm$ 0.0032
	GRAND	0.6715 $\pm$ 0.0490	-0.0942 $\pm$ 0.3897	0.6602 $\pm$ 0.1393
	GraphCON	0.0964 $\pm$ 0.0620	-1.3836 $\pm$ 0.0092	0.6833 $\pm$ 0.0074
	ADGN	-0.5188 $\pm$ 0.1812	-3.2417 $\pm$ 0.0751	0.4296 $\pm$ 0.1003
	Ours	-0.5249 $\pm$ 0.0155	-3.2370 $\pm$ 0.0834	0.4094 $\pm$ 0.0764
Table 3: Performance of different versions of SWAN on Graph Property Prediction and LRGB tasks. The first, second, and third best results for each task are color-coded.	SWAN	-0.5249 $\pm$ 0.0155	-3.2370 $\pm$ 0.0834	0.4094 $\pm$ 0.0764
	SWAN-LEARN	-0.5981 $\pm$ 0.1145	-3.5425 $\pm$ 0.0830	-0.0739 $\pm$ 0.2190
Table 4: Performance of different versions of SWAN on Graph Property Prediction and LRGB tasks. The first, second, and third best results for each task are color-coded.	Model	Peptides-func AP $\uparrow$	Peptides-struct MAE $\downarrow$	Pascal VOC-sp F1 $\uparrow$
	MPNNs			
	GCN	59.30 $\pm$ 0.23	0.3496 $\pm$ 0.0013	12.68 $\pm$ 0.60
	GINE	54.98 $\pm$ 0.79	0.3547 $\pm$ 0.0045	12.65 $\pm$ 0.76
	GCNII	55.43 $\pm$ 0.78	0.3471 $\pm$ 0.0010	16.98 $\pm$ 0.80
	GatedGCN	58.64 $\pm$ 0.77	0.3420 $\pm$ 0.0013	28.73 $\pm$ 2.19
Table 5: Performance of different versions of SWAN on Graph Property Prediction and LRGB tasks. The first, second, and third best results for each task are color-coded.	Transformers			
	Transformer+LapPE	63.26 $\pm$ 1.26	0.2529 $\pm$ 0.0016	26.94 $\pm$ 0.98
	SAN+LapPE	63.84 $\pm$ 1.21	0.2683 $\pm$ 0.0043	32.30 $\pm$ 0.39
	GraphGPS+LapPE	65.35 $\pm$ 0.41	0.2500 $\pm$ 0.0005	37.48 $\pm$ 1.09
Table 6: Performance of different versions of SWAN on Graph Property Prediction and LRGB tasks. The first, second, and third best results for each task are color-coded.	DE-GNNs			
	GRAND	57.89 $\pm$ 0.62	0.3418 $\pm$ 0.0015	19.18 $\pm$ 0.97
	GraphCON	60.22 $\pm$ 0.68	0.2778 $\pm$ 0.0018	21.08 $\pm$ 0.91
	ADGN	59.75 $\pm$ 0.44	0.2874 $\pm$ 0.0021	23.49 $\pm$ 0.54
Table 7: Performance of different versions of SWAN on Graph Property Prediction and LRGB tasks. The first, second, and third best results for each task are color-coded.	Ours			
	SWAN	63.13 $\pm$ 0.46	0.2571 $\pm$ 0.0018	27.96 $\pm$ 0.48
	SWAN-LEARN	67.51 $\pm$ 0.39	0.2485 $\pm$ 0.0009	31.92 $\pm$ 2.50

Dataset $\downarrow$ / Model $\rightarrow$	SWAN $\beta=0$	SWAN-NE	SWAN-NE-LEARN	SWAN	SWAN-LEARN
Diam. ($\log_{10}$ (MSE) $\downarrow$)	-0.3882 $\pm$ 0.0610	-0.5497 $\pm$ 0.0766	-0.5631 $\pm$ 0.0694	-0.5249 $\pm$ 0.0155	-0.5981 $\pm$ 0.1145
SSSP ($\log_{10}$ (MSE) $\downarrow$)	-3.2061 $\pm$ 0.0416	-3.1913 $\pm$ 0.0762	-3.5296 $\pm$ 0.0831	-3.2370 $\pm$ 0.0834	-3.5425 $\pm$ 0.0830
Ecc. ($\log_{10}$ (MSE) $\downarrow$)	0.5573 $\pm$ 0.0247	0.3792 $\pm$ 0.1514	0.1317 $\pm$ 0.1253	0.4094 $\pm$ 0.0764	-0.0739 $\pm$ 0.2190
Peptides-func (AP $\uparrow$)	61.95 $\pm$ 0.67	61.19 $\pm$ 0.37	62.49 $\pm$ 0.51	63.13 $\pm$ 0.46	67.51 $\pm$ 0.39
Peptides-struct (MAE $\downarrow$)	0.2703 $\pm$ 0.0023	0.2672 $\pm$ 0.0012	0.2606 $\pm$ 0.0007	0.2571 $\pm$ 0.0018	0.2485 $\pm$ 0.0009

Table 3: Performance of different versions of SWAN on Graph Property Prediction and LRGB tasks. The first, second, and third best results for each task are color-coded.

4. Considerations and Conclusions

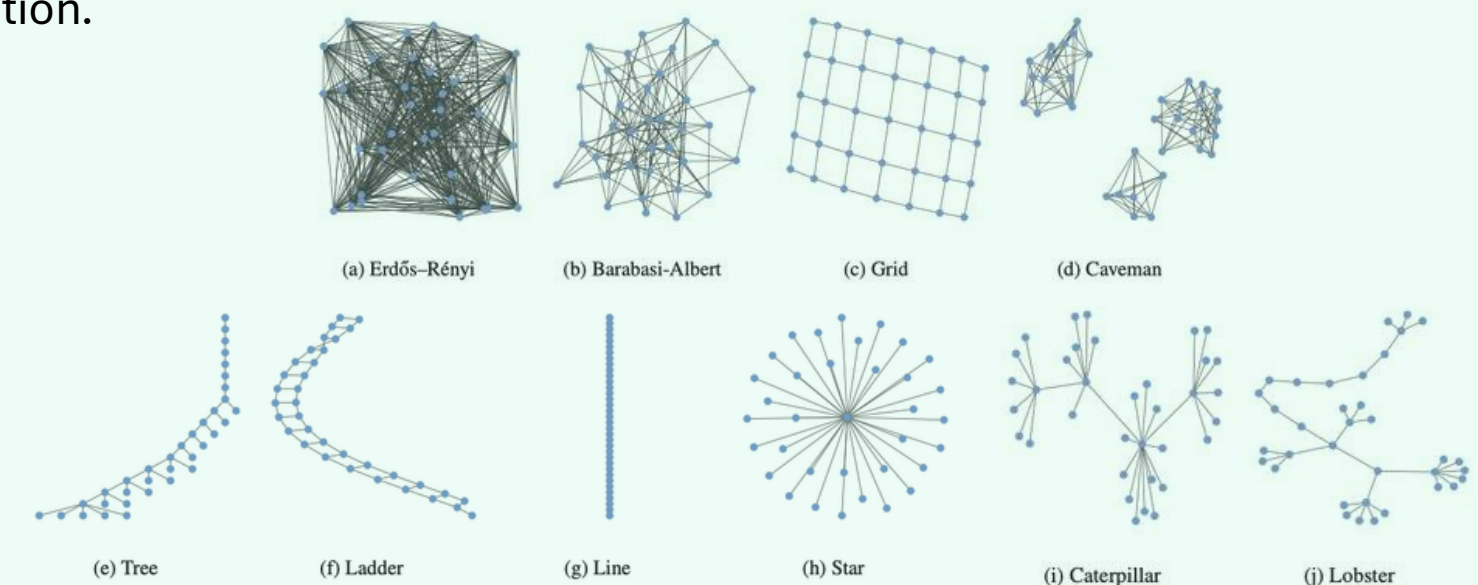
Local and Global Non-Dissipativity

Relaxing structural constraints degrades performance: Non-Enforced variants (SWAN-NE, SWAN-LEARN-NE) and SWAN $\beta=0$ score significantly lower across tasks.

The key winning factor is the global and local non-dissipativity.

Robustness Across Graph Geometries

The architecture was evaluated on diverse 35-node geometries. From dense structures to challenging topologies (chains, trees, grids), it proves highly resilient for long-range propagation.



While traditional models fail on chains (Line, Ladder) due to oversquashing, SWAN successfully transfers long-range information across any topology.

The Efficiency-Accuracy Balance

Scalability Edge: When computational complexity is not a constraint, models such as Graph Transformers can achieve higher performance. However, their quadratic complexity $O(|V|^2)$ can make them impractical due to Out-of-Memory errors on larger graphs.

Linear Complexity: SWAN operates at an efficient $O(|V|+|E|)$ cost, matching standard MPNNs in runtime while significantly improving accuracy.

Optimal Balance: While not a universal solution, SWAN offers the best trade-off between theoretical soundness, predictive performance, and computational efficiency for long-range interactions on real-world graphs.

A General Framework for Non-Dissipative DE-GNNs

SWAN introduces a novel DE-GNN paradigm based on non-dissipativity. Enforced via space-weight antisymmetry, this inductive bias prevents information decay and establishes a general framework for long-range graph modeling.